

Examining VR Technologies for Immersive Lighting Performance Simulation and Visualisation in Building Design and Analysis

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ABSTRACT

This paper presents IVR-LPSim: An Immersive Virtual Reality (VR) Lighting Performance Simulator (LPS) capable of real-time indoor artificial illuminance, glare, daylighting and energy consumption calculation and visualisation. Current LPS's consist of traditional exocentric CAD-based interfaces built for desktop-based PC displays. Lighting performance data is represented using reports, numerical datasets, two-dimensional graphs, charts and static visualisations. Additionally, PC-based LPS's do not provide real-time simulation support, creating a gap between the design and analysis process. As an alternative to current approaches, we present a novel LPS that leverages VR technologies to support an egocentric approach for lighting performance design and analysis. Three immersive lighting performance visualisations are presented to facilitate the representation of lighting performance data within IVR-LPSim.

Index Terms: Virtual Reality—Immersive Visualisation—Building Information Modelling—Lighting Performance Simulation Lighting Design—Lighting Analysis—Illuminance—Glare;

1 INTRODUCTION

Building Performance Simulators (BPS) play a pivotal role in the architectural design process of a typical building and construction project life-cycle. BPS applications are designed to analyse and assess the performance of building design features through accurate simulations of data. Using this data, designers can better understand how effectively and efficiently building design features will perform before beginning the construction process. As a result, a BPS system is critical to the building project as it can allow designers to identify performance issues and mitigate potential construction reworks from occurring later in the project life-cycle, saving significant time, costs, and resources [8]. One building design feature that is fundamental to all building designs is lighting. Lighting is one of the leading contributors to electricity usage within Australia, accounting for up to 40% of electricity usage within commercial premises and 10% of household electricity budgets [1]. Previous research has also demonstrated that poor lighting performance can negatively impact building occupants' physical and mental health [5].

Designers utilise modern Lighting Performance Simulators (LPS) such as DIALux¹ to ensure adequate lighting performance of buildings before commencing construction. How designers interpret and understand the data produced by LPS is essential to making informed decisions during the lighting design and analysis process for buildings. Typical approaches to represent lighting performance in modern LPS consist of numerical datasets, reports, two-dimensional graphs, charts and static visualisations. Additionally, these LPS's are

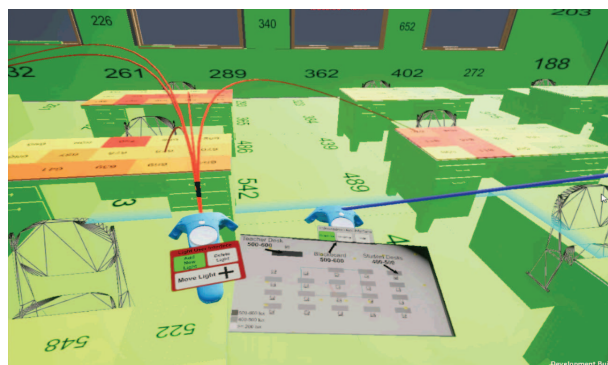


Figure 1: The VR bendrays visualisation for overlit desks and an underlit blackboard within a virtual classroom.

explicitly built for traditional PC two-dimensional displays consisting of standard CAD-based user interfaces. LPS's such as DIALux consists of two visualisation modes of perspective: two-dimensional top-down views, where surfaces are parallel to the floor. This visualisation mode does not easily convey the impact of lighting performance within three-dimensional environments, such as uneven heights of lighting fixtures, overhead cabinets, and interior partitions. To address this, DIALux provides a three-dimensional viewing mode, which allows for a perspective-based approach for lighting design and analysis. However, the lack of spatial awareness still makes it challenging to interpret the impact of lighting performance on surfaces and occupants within the three-dimensional context of the building [2].

To address these issues, we propose IVR-LPSim: An Immersive Virtual Reality Lighting Performance Simulator. The simulator supports real-time lighting simulation to provide live lighting performance feedback for simultaneous lighting design and analysis from an immersive first-person perspective within the building. IVR-LPSim supports various simulation features, including indoor artificial illuminance simulation, glare simulation, daylighting simulation, and energy consumption calculation. Three 3D visualisation techniques suitable for the egocentric nature of VR are integrated into IVR-LPSim to represent real-time lighting feedback. This paper presents an overview of IVR-LPSim which describes the simulation and visualisation components integrated into the system.

2 SYSTEM OVERVIEW

This section provides an overview of the two primary simulation components - artificial illuminance and glare simulation, and the three 3D visualisations - Heatmaps, Bendrays, and Animated Arrows, that are utilised to visualise the simulated lighting performance data.

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¹<https://www.dialux.com/en-GB/>

2.1 Indoor Artificial Illuminance Simulation

The following equations are used to calculate the lux values on a particular surface within the virtual environment: The Apex Angle α is calculated as follows:

$$\alpha = 2\pi(1 - \cos \frac{2\theta}{2}) \quad (1)$$

Where θ is equal to the cone apex angle in degrees. The Candela value (I_v) is calculated as follows: Where L is equal to the lumens of the luminaire. The lux value (E_v) is calculated as follows: Where D is equal to the distance from the luminaire to the centre point of a submesh.

$$I_v = \frac{L}{2\pi(1 - \cos 1/2\alpha)} \quad (2) \quad E_v = \frac{I_v}{D^2} \quad (3)$$

2.2 Glare Simulation

The Unified Glare Rating (UGR) [3] illustrated in Equation 4 is utilised in IVR-LPSim to calculate glare discomfort [7]. The UGR equation is used to determine the likelihood of a luminaire causing glare discomfort to surrounding objects. Outputted UGR values can range from less than 10, being imperceptible, to greater than 28, being uncomfortable. The following equation is used to calculate the UGR values on a particular surface within the virtual environment:

$$UGR = 8 \log[0.25/Lb \sum (L^2 \omega / p^2)] \quad (4)$$

Where Lb is the background luminance, L is the luminance of the luminous parts of each luminaire in the direction of the observer's eye, ω is the solid angle of the luminous parts of each luminaire at the observer's eye, and p is a modified version of the positional index for each luminaire originally proposed by Iwata and Tokura [4]. The position index is calculated by:

$$p = \begin{cases} 1 + (0.8R)/D, & \text{if } R < 0.6D; \\ 1 + (1.2R)/D, & \text{else } R \geq 0.6D. \end{cases} \quad R = \sqrt{Z^2 + Y^2} \quad (6)$$

Where D is the distance eye to plane of the source in the view direction, Y is the vertical distance between the source and eye, and Z is the horizontal distance between the source and eye.

2.3 Visualisations

For designers to understand and interpret simulated lighting information within a LPS, a data representation method is required to provide feedback. This section presents three easy-to-use 3D visualisations to represent lighting performance data to support users in making informed decisions about the lighting layout of a building.

2.3.1 3D Heatmaps

3D Heatmaps provide designers with a colour-coded representation of lighting performance. Our implementation consists of a three-colour gradient scale, where blue colours represent underlit areas, red colours represent overlit areas, and green colours represent within specification lit areas. The 3D Heatmap consists of an incremental transition of colours, meaning that a lighter or darker shade of a specific colour will vary based on the lux or UGR values of the surface, resulting in a natural blend of colours on surfaces.

2.3.2 Bendrays

A limitation of 3D Heatmaps is the inability to perform lighting analysis for surfaces beyond a user's FOV. The Bendrays Visualisation aims to address this limitation by providing visual cues that guide users towards surfaces that contain lighting design issues that are beyond their FOV. The visualisation produces a set of curved rays based on a bezier curve algorithm [6, 9] that curve from the user's

controllers to surfaces in the virtual environment that are underlit, overlit, or producing glare, see Figure 1. For the indoor artificial lux simulation, a set of ten blue rays are cast from the right controller towards the underlit surfaces, and a set of ten red rays from the left controller are cast to the overlit surfaces. A limit of ten rays at a time was chosen to prevent the visualisation from distracting the user or occluding their view; however, these parameters can be adjusted for more complex environments.

2.3.3 Animated Arrows

The Animated Arrow visualisation was designed to assist the user with identifying luminaires in the virtual environment that are causing a particular surface to be overlit or producing glare. The visualisation emits virtual arrow cues from a luminaire that is causing the lighting issue. The animation of the arrow cues is based on the light ray trajectory from the luminaire to the surface. When multiple luminaires are impacting the lighting performance of a specific overlit surface, the luminaire producing the highest lux value to that surface is selected. The process may be repeated in cases of multiple lighting issues. This process allows designers to use Animated Arrows to understand and detect specific luminaires causing a specific surface within the virtual environment to be overlit or have glare issues.

3 CONCLUSION

This paper presented IVR-LPSim: An Immersive Virtual Reality (VR) Lighting Performance Simulator (LPS) that is capable of real-time artificial illuminance, glare, daylight and energy consumption simulation. The three new VR interactive visualisations 3D Heatmaps, Bendrays, and Animated Arrows are introduced to provide live lighting performance feedback for simultaneous lighting design and analysis. Our previous studies have established that IVR-LPSim can significantly outperform DIALux in regard to identifying and fixing common interior lighting design issues. Future directions of this work will focus on expanding the capabilities of IVR-LPSim by conducting user studies to evaluate the daylighting and energy consumption simulation features. Future investigations will additionally aim to gain a deeper understanding of how IVR-LPSim can improve the design workflow for professional architects.

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